Lab No 03

Task - 1

```
1- # include <iostream>
2- Using namespace std;
3- int main ( ) {
4-     int age ;
5-     double pay;
6-     Char Section;
7- Cout <<"Enter age pay and section:
8- Cin >> age >> pay >> section;
9- Cout << "Age; " << age << "; pay" << "; Section" << Section << end1;
10- Return 0
11- }
```

Output : Task - 1

```
Enter age pay and section 19 25000.5 E
Age : 19 pay 25000.5 Section E.
```

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main.cpp

Run ▶

```cpp
#include <iostream>
using namespace std;
int main() {
    int age;
    double pay;
    char section;

    cout << "Enter age, pay and section: ";
    cin >> age >> pay >> section; // single cin statement

    cout << "Age: " << age << ", Pay: " << pay << ", Section: " <<
section;
    return 0;
}
```

Output

```
Enter age, pay and section:
```

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Task - 2

```
1- # include <iostream>
2- Using namespace Std;
3- int main ( ) {

4- Cout << 28/4 - 2 << end1; // 5
5- Cout << 6 +12*2-8 << end1; // 22
6- Cout << 4+8 * 2 << end 1; // 20
7- Cout << 6+17% 3-2 << end1; // 46
8- Cout << (8+7) * 2 << end1 ;// 30
9- Cout << (16+7) % 2 - 1. << end1; // 0
10- Cout << 12/ (10 -6) << end1 ;// 3
11- Cout << (19-3) * (2+2) /4 << end1; // 16
12- return 0;
13- }
```

Output: Task - 2

```
5
22
20
6
46
30
0
3
16
```

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main.cpp

Run ▶

```cpp
#include <iostream>
using namespace std;
int main() {
    cout << 28 / 4 - 2 << endl;
    cout << 6 + 12 * 2 - 8 << endl;
    cout << 4 + 8 * 2 << endl;
    cout << 6 + 17 % 3 - 2 << endl;
    cout << 2 + 22 * (9 - 7) << endl;
    cout << (8 + 7) * 2 << endl;
    cout << (16 + 7) % 2 - 1 << endl;
    cout << 12 / (10 - 6) << endl;
    cout << (19 - 3) * (2 + 2) / 4 << endl;
    return 0;
}
```

Output

```
5
22
20
6
46
30
0
3
16
```

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Task-3

1- // Weight = mass * units;

2- // Answer: option C is most accurate

3- // units (int) is promoted to float, mass remain float.

4- // the result of mass * unit is float, which is then Promoted to double for weight.

Output: Task-3

Answer: option C

Using (int) -> promoted to float, mass is float result is promoted to double for weight variable.

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Task-4

1- int a, b = 2;

2- float c = 4.2;

3- a = b * c; // 2 * 4.2 = 8.4 -> truncated to 8

4- // Answer : B. 8

Output : Task-4

8

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Task-5

```
1. # include <iostream>
2. Using namespace std;
3. int main () {
4.     int qty = 10;
5.     int saleReps = 3;
6.     double UnitEach;
7. // Wrong : UnitEach = qty / SalesReps.
8. // correct with type cast :
9. UnitEach = static_cast<double> (qty) SalesReps;
10. // other valid forms :
11. // UnitEach = (double) qty / salesReps;
12. // UnitEach = (double) (qty) / SalesReps:
13. Cout << unitsEach << end1;
14. Return 0;
15. }
```

Output Task-5

3.33333

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main.cpp

Run ▶

Output

```cpp
#include <iostream>
using namespace std;
int main() {
    int qty = 10, salesReps = 3;
    double unitsEach;

    // Correct way with type casting
    unitsEach = static_cast<double>(qty) / salesReps;
    // OR unitsEach = (double)qty / salesReps;
    // OR unitsEach = double(qty) / salesReps;

    cout << unitsEach; // 3.33 not 3
    return 0;
}
```

```
3.33333
```

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Task - 6

```
1-  #include <iostream>
2-  #include<cstdlib>
3-  #include<ctime>
4- Using namespace std;
5- int main ( ) {
6-   Srand(time(0));
7-   int num1= rand( )%900+100; //100 to 999 Lik 247
     int num2= rand( )%900+100; // Like 129
8-  Cout <<" "<< num1 << end1;
9-  Cout <<"+ "<< num2 << end1;
10- Cout <<"....." << end1;
11- Cout<< "\npress tnter When ready to see answer.....";
12- Cin.get( );
13- Cout<<" "<< num1<<end1;
14- Cout <<"+"<< num2<< end1;
15- Cout <<"....." << end1;
16- Cout <<"   "<< num1+num2<< end1;
17- return 0
18- }
```

OutPut: Task-6

```
 247
+129
Press enter When ready to see answer....
 247
+129
 376
```

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main.cpp

Run ▶

```cpp
using namespace std;

int main() {
    srand(time(0)); // random seed

    int num1 = rand() % 900 + 100; // 100 to 999 tak random number
    int num2 = rand() % 900 + 100;

    cout << "  " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "--------" << endl;

    cout << "\nPress Enter to see answer...";
    cin.get(); // pause

    cout << "  " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "--------" << endl;
    cout << "  " << num1 + num2 << endl;

    return 0;
}
```

Output

```
  943
+ 747
--------

Press Enter to see answer...
  943
+ 747
--------
  1690
```

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Home Task-1 Program 1

```
1- #include <iostream>
2- Using namespace Std;
3- int main () {
4- double number, number2, Sum;
5- Cout <<"Enter a number:";
6- Cin>> number1;
7- Cout <<" Enter another number:";
8- Cin >> number2;
9- Sum = number1+number2;
10- Cout<< The sum of the two numbers is" << sum<< end:
11- return 0
12- }
```

Output:

Enter a number: 10

Enter another number: 20

The Sum of the two number is 30

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main.cpp

Run ▶

```cpp
#include <iostream>
using namespace std;

int main() {
    double number1, number2, sum;
    cout << "Enter a number: ";
    cin >> number1; // Error 1: Cin << was wrong, should be cin >>
    cout << "Enter another number: ";
    cin >> number2; // Error 2: cin << was wrong
    sum = number1 + number2; // Error 3: number1+number2=sum is wrong
    cout << "The sum of the two numbers is " << sum; // Error 4: cout << missing, and "" << missing
    return 0;
}
```

Output

```
Enter a number: 10
Enter another number: 20
The sum of the two numbers is 30
```

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Home Task - 1 Program - 2

```
1- # include <iostream>
2- Using namespace std;
3- int main () {
4-    int number1, number2;
5-    float quotient;
6- Cout << "Enter two number and I will divide\n";
7- Cout<< "the first by the Second for you.\n";
8- Cin >> number1 >> number2;
9- quotient = Static cast <float> (number1) / number2;
10- Cout << quotient << endl;
11- return 0
12- }
```

OutPut:

```
Enter two numbers and I will divide
the first by the Second for you
10 2

5
```

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main.cpp

Run ▶

Output

```cpp
#include <iostream>
using namespace std;

int main() {
    int number1, number2;
    float quotient;
    cout << "Enter two numbers and I will divide\n";
    cout << "the first by the second for you.\n";
    cin >> number1 >> number2; // Error: comma not allowed, use >>
    quotient = static_cast<float>(number1) / number2; // Error: float<static_cast> syntax wrong
    cout << quotient; // Error: missing ; and missing }
    return 0;
}
```

```
Enter two numbers and I will divide
the first by the second for you.
```

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Home Task - 2

```
1- # include <iostream>
2- Using namespace std;
3- int main () {
4-    int V1 = 28, V2 = 32, V3 = 37 V4 = 24 V5 = 33;
5-    int Sum = V1 + V2 + V3 + V4 + V5
6-    double average = Sum / 5.0
7-    Cout << "sum = " << Sum << end 1;
8-    Cout << Average = " << average << end 1;
9-    return 0
10-  }
```

Output:-

Sum = 154

Average = 30.8

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main.cpp

Run ▶

```cpp
#include <iostream>
using namespace std;
int main() {
    int a=28, b=32, c=37, d=24, e=33;
    int sum;
    double average;

    sum = a + b + c + d + e;
    average = sum / 5.0; // 5.0 se divide taake decimal aaye

    cout << "Sum = " << sum << endl; // Sum = 154
    cout << "Average = " << average << endl; // Average = 30.8

    return 0;
}
```

Output

```
Sum = 154
Average = 30.8
```

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